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Bhadla Solar Park, Rajasthan

Bhadla solar park is a 2.25GW solar complex being developed in Bhadla village in Jodhpur district of Rajasthan, India.

Project Type	Location	Capacity	Start of Construction	Expected Investment
Solar park	Jodhpur District, Rajasthan, India	2.25GW	July 2015	Rs98.5bn ($1.4bn)

Expand

Ecoppia will install approximately 2,000 robotic solar panel cleaning equipment at the solar park. Image courtesy of Ecoppia.

Bhadla solar park is a 2.25GW solar complex being developed in Bhadla village in Jodhpur district of Rajasthan, India.

The project is being developed in four phases, with Rajasthan Solar Park Development Company Limited (RSPDCL) developing the first two phases, Saurya Urja Company of Rajasthan developing phase three, and Adani Renewable Energy Park Rajasthan developing phase four.

The total estimated investment on the project is Rs98.5bn ($1.4bn).

The solar park construction was started in July 2015 and the first phase was commissioned in October 2018. The second phase is expected to be commissioned in April 2019, while phases three and four are expected to be commissioned by March 2019.

The project is being developed under the Ministry of New & Renewable Energy (MNRE) scheme for the development of solar parks and ultra-mega solar power projects and is in line with the Rajasthan Solar Energy Policy, 2011.

Bhadla solar park location

Bhadla solar park spans a total area of 5,783ha in Bhadla village, which is located 220km away from Jodhpur on Bap-Bhadla road.

Rajasthan Renewable Energy Corporation (RREC) has allotted the land for the execution of the project in co-ordination with the state government. RSPDCL leased 1,800ha of land to develop phase two of the project.

Rajasthan has the highest solar irradiation of 5.72kWh/m²/day making it a perfect site for solar park development. The country also has the advantage of maximum solar power potential of 142GW in the country with vast unused barren and affordable land.

Bhadla solar park details

The first phase of the solar park has seven solar power plants with a combined capacity of 75MW, while phase two has ten solar power plants with a combined capacity of 680MW.

Phases three and four will have ten solar power plants each, with combined capacities of 1,000MW and 500MW respectively.

National Thermal Power Corporation (NTPC) and Solar Energy Corporation of India (SECI) are responsible for assigning developers for the solar power plants. The two organizations have signed 25-year power purchase agreement with developers.

The solar power plants under the third phase are being developed by Hero Future Energies (300MW), Softbank Group (200MW), ACME Solar (200MW), and SB Energy (300MW).

Azure Power (200MW), ReNew Solar Power (50MW), Phelan Energy Group (50MW), Avaada Power (100MW), and SB Energy (100MW) are undertaking development of the solar power plants under the fourth phase.

Power transmission from Bhadla solar park

The power evacuation system of the solar park will be developed by Powergrid Corporation of India (PGCIL) and state-owned transmission company TRANSCO and delivered to the state electricity boards.

PGCIL will establish 765/400/220kV grid substations and pooling stations at Bhadla. A transmission system with 765kV and 400kV double-circuit line along with two 400kV and four 220kV line bays for solar parks interconnection will also be laid.

TRANSCO will also develop 400/220/132kV grid substations for power evacuation.

Financing

Construction of the new transmission lines from the solar park to the national grid is being financed under the Rajasthan Renewable Energy Transmission Investment Programme (RRETIP), which is co-ordinated by the Government of India, the Asian Development Bank (ADB), and the Clean Technology Fund.

RRETIP has rendered technical assistance to plan infrastructure and implement phase two of the project and to plan and review phases three and four.

Contractors involved

Rays Power Experts was the engineering, procurement and construction (EPC) contractor for the development of 140MW capacity in phase two of the solar park.

Vikram Solar was the EPC contractor for a 130MW solar power plant in phase two.

Larsen & Toubro secured the contract to build a 100MW solar power plant in phase three from SB Energy in February 2018.

Ecoppia signed a contract with SB Energy to install 2,000 robotic solar panel cleaning equipment across five sites in phases three and four in June 2018.

Meghraj Capital Advisors Private Limited, RangSutra, and Urmul Marusthali Bunkar Vikas Samiti provided technical assistance during project development.

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